TODAY'S LESSON

THE HISTORY OF AI

From an impossible dream to machines that can do the impossible.



Five Acts, One Question

I	**The Dream**	Two moments in the 1950s when people first dared to ask: can a machine think?
II	**Two Winters**	Big hype, a hard crash, a second act of hype, a second crash. Twice.
III	**The Turning Point**	What quietly changed in the background while everyone had given up.
IV	**Three Breakthroughs**	Learning, seeing, and reading — the three tricks behind everything AI does today.
V	**The Verdict**	We return to today's question, armed with the whole story.

Next: we rewind to 1950, before there was any AI to argue about.

1950: A Question, Not an Answer

1950

Alan Turing
Computing Machinery
and Intelligence

Alan Turing doesn't try to answer "can machines think?" — he thinks the question itself is too vague to argue about. Instead he proposes a test: **the Imitation Game.** If a human judge, chatting blind with a machine and a person, can't reliably tell which is which, the machine passes.

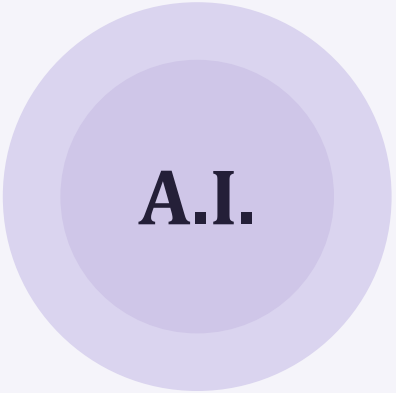
Why this matters: Turing sidesteps a philosophy debate about consciousness and replaces it with something you can actually run an experiment on. That reframing — judge the behavior, not the "soul" — is still how most AI is evaluated today.

Next: the idea needs a name, and a room full of researchers is about to give it one.

1956: The Field Gets a Name

Six years later, a small group gathers for a summer workshop at Dartmouth College. Their proposal makes a bold claim: *every aspect of learning or intelligence can, in principle, be described precisely enough that a machine can simulate it.*

John McCarthy coins a name for this new field: **"Artificial Intelligence."** The organizers guess the whole workshop's problems could be cracked in about two months, with a handful of people. It will take a lot longer than that.



Dartmouth Summer Research Project, 1956

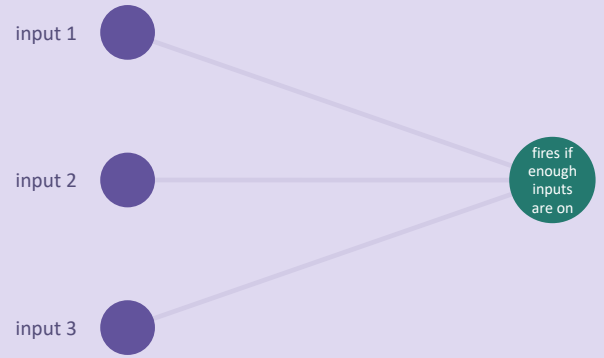
Next: before you can dream about a thinking machine, someone has to try building a brain out of wires.

Building a Brain Out of Switches

Rewind slightly: seven years before Turing's question, two researchers had already sketched the first rough blueprint for an artificial brain.

1943 — The First Model Neuron

Warren McCulloch and Walter Pitts publish a mathematical model of a neuron: a simple switch that "fires" only once enough of its inputs are active. It's not built from anything biological — just logic and arithmetic — but it proves a real neuron's behavior can be captured in math.



A single artificial neuron: simple, but it's the direct ancestor of every network in this deck.

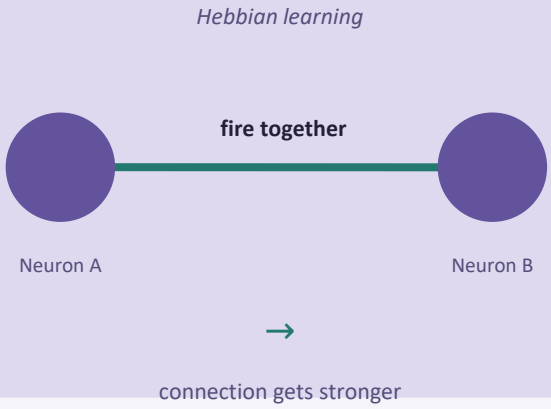
Next: the idea leaves the whiteboard — someone actually builds one, and the press goes wild.

1949: What If Connections Could Learn?

McCulloch and Pitts had shown how a neuron could compute. Donald Hebb asked the next question: how could a network of neurons change with experience?

1949 — Donald Hebb

Hebb proposed that connections between neurons could become stronger when the neurons repeatedly became active together. In simple terms: experience could change the strength of connections inside a network.



Next: Rosenblatt takes the idea of adjustable connections and turns it into a machine that learns from mistakes.

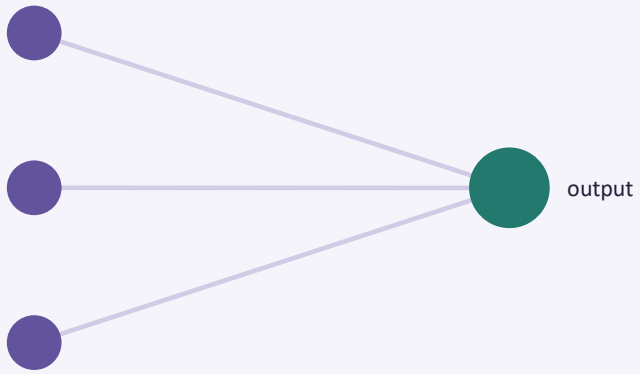
THE FIRST ATTEMPT

1958: A Machine That Learns — and a Media Frenzy

Frank Rosenblatt builds the **Perceptron**: a machine that adjusts its own internal settings after each mistake, gradually getting better at telling two kinds of things apart. It's the first machine that genuinely learns from examples instead of being explicitly programmed.

The hype: newspapers at the unveiling describe a coming machine that could walk, talk, see, write, and even reproduce itself — all within Rosenblatt's own lifetime. The Perceptron could actually do none of that yet.

A perceptron: inputs → weights → decision



A weight adjusts after every mistake — this is the whole learning rule.

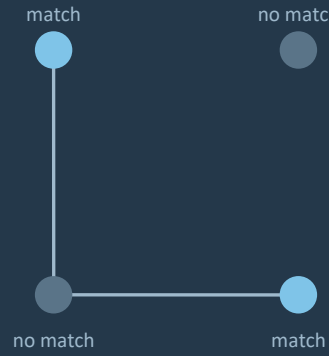
Next: someone checks the math behind the hype — and the answer isn't pretty.

1969: The Hype Meets the Math

Marvin Minsky and Seymour Papert — *one of them present at Dartmouth thirteen years earlier* — publish a book proving something uncomfortable: a single-layer Perceptron can't even solve XOR, a problem a child can learn in seconds.

The fallout: funding agencies read it as proof that neural networks were a dead end. Combined with a skeptical 1973 government report in the UK, money for AI research dries up almost overnight. This is **AI Winter I.**

The problem a perceptron CAN'T learn (XOR)

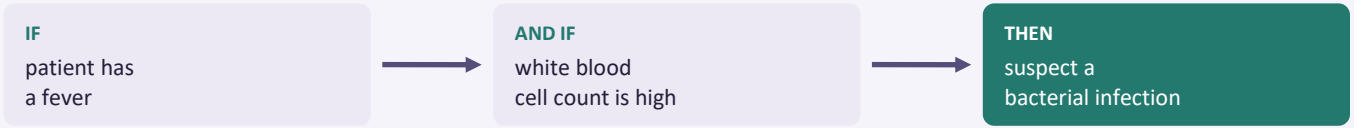


No single straight line can separate the two groups — you'd need to bend the line. A perceptron can only draw straight lines.

Next: researchers try a completely different strategy — teach the machine rules instead of letting it learn.

If This, Then That: The Rule-Based Era

Through the 1970s and 80s, researchers try a different approach entirely: stop trying to make machines learn, and instead **hand-write the rules** a human expert would use. These "expert systems" ask a chain of IF-THEN questions to reach a conclusion.



This example mimics MYCIN, a real 1970s expert system for diagnosing infections.

MYCIN

Diagnosed bacterial infections about as well as human specialists

XCON

Configured computer orders at Digital Equipment Corp, saving millions a year

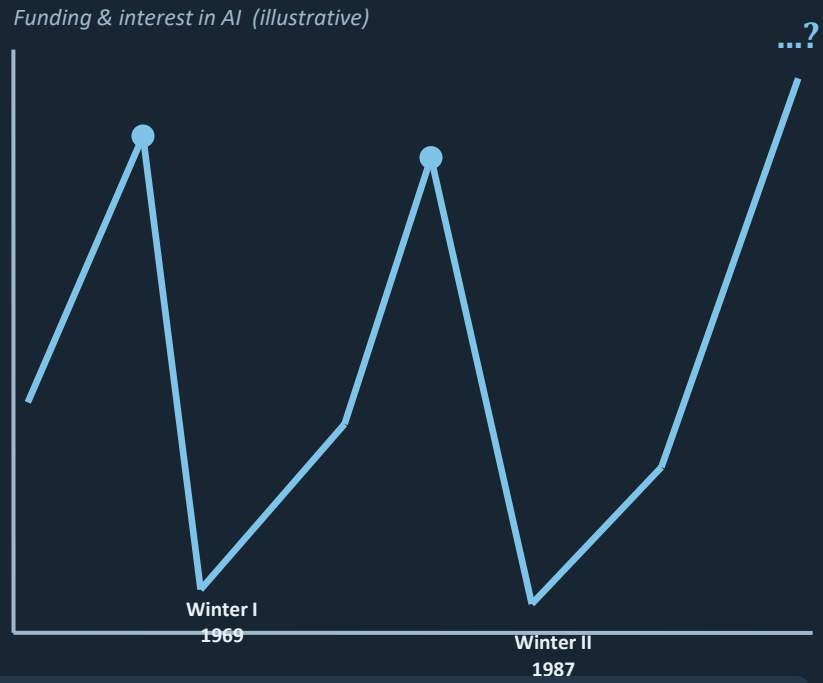
National bets

Japan launches the Fifth Generation Computer Project, betting big on this approach

1987: History Repeats Itself

Expert systems turn out to be **brittle**: ask them anything outside their exact rules and they fail completely, with no ability to guess or generalize. Every new rule has to be hand-coded by an expensive specialist.

The trigger: in 1987, the market for specialized "LISP machine" hardware — built specifically to run these systems — collapses almost overnight as cheaper general-purpose computers catch up and overtake them. Funding and confidence crash for a second time.



Next: twice burned — so what was actually different the third time around?

What Was Different This Time?

While almost everyone had stopped paying attention, three separate trends were quietly building in the background — and by the 2000s, they finally lined up at once.

1

DATA

The internet quietly generated oceans of labeled examples — photos, text, clicks — the raw material learning needs.

2

COMPUTE

GPUs, originally built to render video game graphics, turn out to be exactly the right shape of hardware for the math neural networks need.

3

ALGORITHMS

Decades-old ideas like backpropagation had been sitting on the shelf the whole time — they just needed enough data and compute to finally prove themselves.

Next: with all three ingredients finally in place, the first real breakthrough arrives — teaching a machine to actually learn from its mistakes.

Rolling Downhill: Gradient Descent

A neural network starts by guessing — badly. Gradient descent is the trick that lets it improve: measure how wrong the guess was (the **error**), then nudge every internal setting slightly in the direction that reduces it. Repeat this millions of times, and the network "rolls downhill" toward better answers.

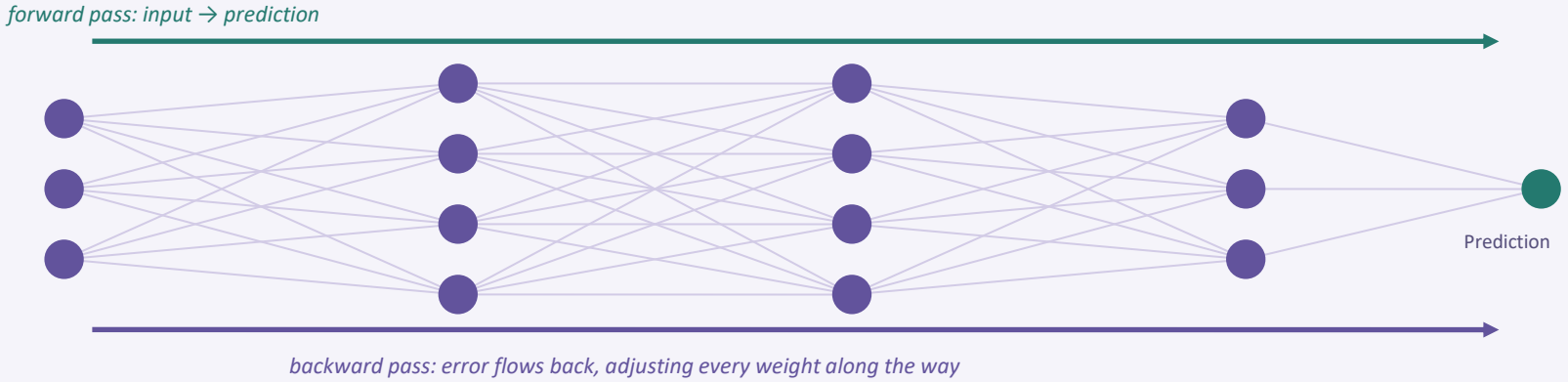


Think of it like: you're blindfolded on a hillside and want to reach the lowest point. You can't see the whole landscape, but you can feel which direction slopes downward under your feet — so you take a small step that way, and repeat.

Next: one problem — this only worked well for shallow networks. A second idea makes it work for deep ones too.

1986: Making Gradient Descent Scale

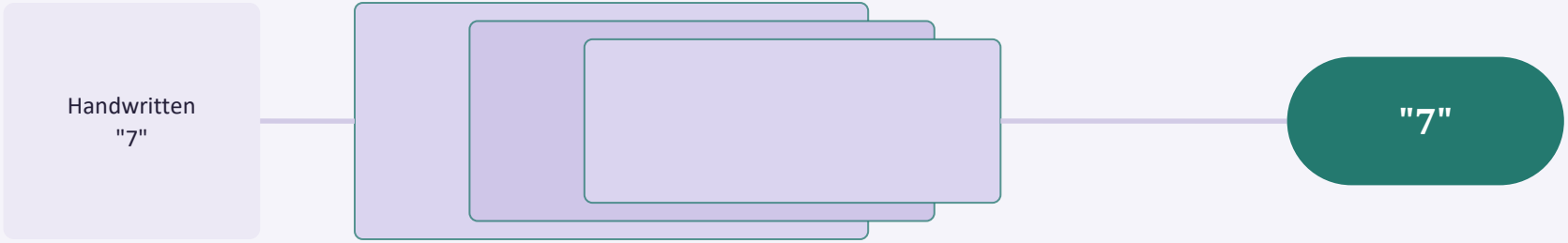
Gradient descent needs to know which direction is "downhill" for every single setting in the network — and a deep network can have millions of them. **Backpropagation**, popularized by Rumelhart, Hinton, and Williams, is an efficient way to compute all of those directions at once by working backward from the error.



Next: with learning finally solved, the next question is: learning from what? Let's teach a machine to see.

1989: Teaching a Machine What a Digit Looks Like

Yann LeCun builds a **Convolutional Neural Network (CNN)** that reads handwritten digits — useful for sorting mail and reading checks. It's inspired by how our own visual cortex works: simple edge-detectors feed into detectors for shapes, which feed into detectors for whole objects.

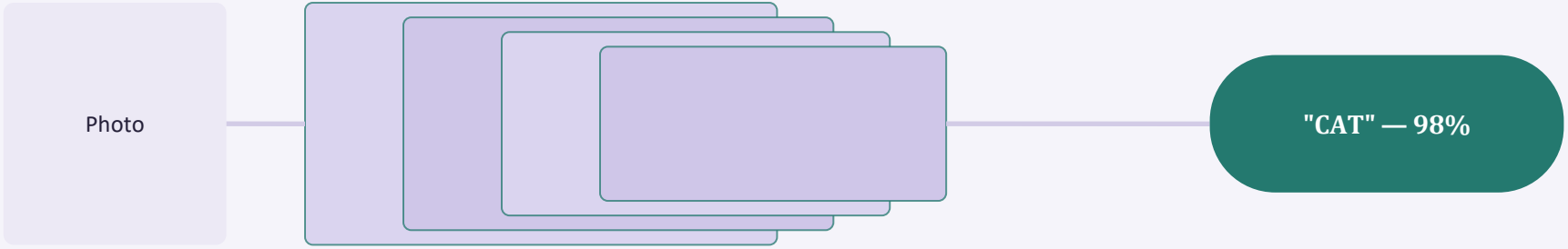


Edge detectors → Stroke shapes → Whole digit

Next: this idea works, but stays a niche curiosity for over two decades — until one competition changes everything.

2012: The Competition That Changed Everything

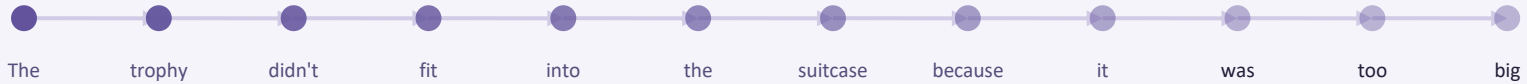
Every year, the **ImageNet competition** challenges teams to correctly label over a million photos into a thousand categories. In 2012, a CNN called **AlexNet** enters and beats the field by a wide margin — far more than anyone expected from a single year's improvement.



Next: this AlexNet moment is the spark for the modern deep learning boom — but sight is only one sense. Language is next.

The Old Way: One Word at a Time

Before 2017, machines read language the way you'd read through a keyhole: one word at a time, left to right, trying to remember everything important that came before. Over a long sentence, early details fade — a real limitation, not just a metaphor.



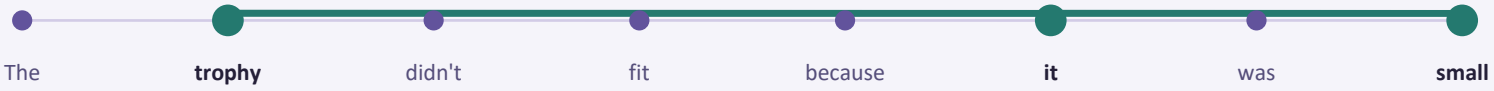
By the time the machine reaches "it," the memory of "trophy" vs. "suitcase" has already started to fade — so which one does "it" refer to?

The fix isn't a bigger memory. It's a completely different idea: let the machine look at every word in the sentence at once.

2017: Attention Changes Everything

A Google research paper titled *"Attention Is All You Need"* introduces the **Transformer**: a model that looks at an entire sentence at once and learns directly which words matter most to each other — no fading memory required.

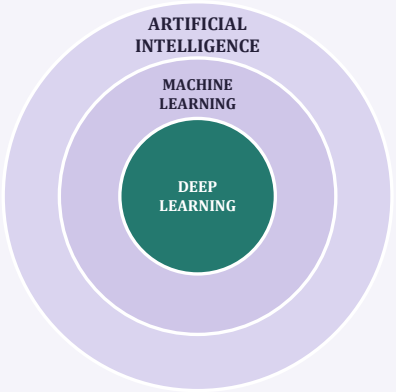
Thicker lines = stronger "attention" the model learned between words — it correctly links "it" to "trophy," not "suitcase."



This one idea is the direct ancestor of GPT, BERT, and virtually every large language model powering today's chatbots.

Next: we now have three breakthroughs — time to zoom out and see how they all fit together.

AI Is Not Just One Thing



Machine Learning — learns from data instead of fixed, hand-written rules.

Deep Learning — many-layered neural networks, trained with backpropagation.

- **Computer Vision**

  The "seeing" breakthrough — recognizing faces, objects, and scenes

- **Natural Language Processing**

  The "reading" breakthrough — understanding and generating language

- **Robotics**

  Sensing and acting in the physical world

- **Generative AI**

  Creating new text, images, music, and code

Next: let's see where all three breakthroughs actually show up in the world around you right now.

AI Today: The Dream, Realized?

Medicine

Reading medical scans and speeding up drug discovery

Science

Predicting protein shapes (AlphaFold) years faster than by hand

Transportation

Real-time perception for self-driving systems

Creativity

Generating art, music, and code from a short prompt

Language

Real-time translation and conversational chatbots

Every single one of these systems learns the exact same three ways you just saw: **gradient descent, convolution, and attention.** Nothing here is magic — it's the same handful of ideas, scaled way up.

Next: back to where we started — is it actually smart?

“We can only see a short distance ahead,
but we can see plenty there that needs to be done.”

— Alan Turing, 1950

From the first artificial neuron to today's AI,

the story is far from over.

QUESTIONS?